

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CONSTRAINT-BASED PROBLEM SOLVING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO5 – Precision (BL4)	Assessment:	Constraint Based Problem Solving
Weightage:	40%	Total Marks:	100
CO5 Statement:	Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Identify the relevant concepts of I/O interfaces, interrupts, DMA, bus arbitration, and high-speed communication protocols.
- Analyze the I/O requirements considering CPU utilization, latency, throughput, bandwidth, and device priorities.
- Apply suitable I/O techniques such as DMA, vectored interrupts, memory-mapped I/O, nested interrupts, and bus arbitration.
- Compute performance measures such as data transfer rate, interrupt latency, CPU utilization, throughput, and transfer time.
- Interpret the results by evaluating CPU efficiency, communication performance, interrupt response, and suitability of the proposed I/O architecture.

QUESTIONS

1. Construct an I/O communication mechanism for a real-time embedded system with the following constraints:
 - CPU utilization must be below 15%
 - Multiple sensors generate interrupts every 1 ms
 - Use vectored interrupt handling
 - Select between memory-mapped and I/O-mapped I/O for optimal latency
2. Design an I/O subsystem under these constraints:
 - High-speed storage requires throughput > 3 GB/s
 - Must use NVMe over PCIe
 - CPU should not handle bulk data transfer
 - Integrate DMA controller with interrupt notification
3. Construct a multi-device communication architecture with constraints:
 - 6 peripherals using USB, 2 high-speed devices using Thunderbolt
 - Avoid interrupt starvation
 - Use hardware and software interrupts
 - Ensure fair bus arbitration
4. Design a data acquisition system with constraints:
 - Continuous data stream at 500 MB/s
 - Minimal CPU intervention required
 - Must compare DMA transfer vs interrupt-driven I/O
 - Use memory-mapped I/O for device registers
5. Construct an interrupt handling mechanism with constraints:
 - Support nested interrupts
 - Maximum interrupt latency < 10 μ s
 - Use vectored interrupts for high-priority devices
 - Use software interrupts for background tasks

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Marks Evaluation:

Q. No	Problem Understanding (25)	Constraint Analysis (25)	Solution Development (25)	Application of Concepts (15)	Justification (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/25	/ 25	/ 15	/ 10	/ 100

Course Coordinator

Faculty Incharge